

Proximity Measure for Categorical Attributes

- ❑ Categorical data, also called nominal attributes
 - ❑ Example: Color (red, yellow, blue, green), profession, etc.
- ❑ Method 1: Simple matching
 - ❑ m : # of matches, p : total # of variables

$$d(i, j) = \frac{p - m}{p}$$

- ❑ Method 2: Use a large number of binary attributes
 - ❑ Creating a new binary attribute for each of the M nominal states

Ordinal Variables

- ❑ An ordinal variable can be discrete or continuous
- ❑ Order is important, e.g., rank (e.g., freshman, sophomore, junior, senior)
- ❑ Can be treated like interval-scaled
 - ❑ Replace *an ordinal variable value* by its rank: $r_{if} \in \{1, \dots, M_f\}$
 - ❑ Map the range of each variable onto $[0, 1]$ by replacing i -th object in the f -th variable by
$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$
 - ❑ Example: freshman: 0; sophomore: 1/3; junior: 2/3; senior 1
 - ❑ Then distance: $d(\text{freshman}, \text{senior}) = 1$, $d(\text{junior}, \text{senior}) = 1/3$
 - ❑ Compute the dissimilarity using methods for interval-scaled variables

Attributes of Mixed Type

- ❑ A dataset may contain all attribute types
 - ❑ Nominal, symmetric binary, asymmetric binary, numeric, and ordinal
- ❑ One may use a weighted formula to combine their effects:

$$d(i, j) = \frac{\sum_{f=1}^p w_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p w_{ij}^{(f)}}$$

- ❑ If f is numeric: Use the normalized distance
- ❑ If f is binary or nominal: $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$; or $d_{ij}^{(f)} = 1$ otherwise
- ❑ If f is ordinal
 - ❑ Compute ranks z_{if} (where $z_{if} = \frac{r_{if} - 1}{M_f - 1}$)
 - ❑ Treat z_{if} as interval-scaled